

Filing Lens

Cited answers from company filings, from a local 3B model (Qwen 2.5, runs on the phone)

Retail investors are handed 100-page 10-Ks. Chatbots answer anyway, with no page reference, and the filing gets uploaded to someone else's server.

FinanceBench (Islam et al., 2023, arXiv:2311.11944): GPT-4-Turbo with a retrieval system **incorrectly answered or refused 81%** of 150 open-book filing questions.

Problem: wrong answers look the same as right ones, and privacy is the price of asking.

Demo

Filing Lens

Local 3B · cited · private

0 doc bytes sent

BESTBUY_2023_10K.pdf

Indexed 75 pages · 388 chunks · stored in this browser only

Switch

Recorded responses (hand-authored test fixture; replace with Kaggle-recorded Qwen2.5-3B outputs)

Test mode: replays recorded responses.

How much cash did operating activities provide in fiscal 2023?

Operating activities provided \$1,824 million of cash in fiscal 2023 p.42.

Confidence 99% Well supported

Calibrated probability the retrieved pages support an answer. Abstains below 50%.

Recorded responses (hand-authored test fixture; replace with Kaggle-recorded Qwen2.5-3B outputs) · 21 ms · citation coverage 100% · sources: p.42

Ask about this filing... Ask

Cited answer + calibrated confidence

Filing Lens

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Page 42 of 75 Page Text X

Consolidated Statements of Cash Flows

in millions

	January 29, 2023	January 29, 2022	January 29, 2021
Operating activities			
Net earnings	\$ 1,418	\$ 2,454	\$ 1,768
Adjustments to reconcile net earnings to total cash provided by operating activities:			
Depreciation and amortization	815	898	838
Restructuring charges	347	234	284
Stock-based compensation	155	141	135
Deferred income taxes	31	14	288
Other, net	12	11	3
Changes in operating assets and liabilities, net of acquired assets and liabilities:			
Accounts receivable	(115)	17	72
Prepaid expenses	(22)	(22)	(42)
Accounts payable	(1,268)	(251)	1,674
Income taxes	38	(156)	173
Other liabilities	(455)	379	488
Total cash provided by operating activities	\$ 1,824	\$ 3,152	\$ 4,957
Investing activities			
Acquisitions, net of cash acquired	(835)	(737)	(713)
Acquisitions of property and equipment, net of cash acquired	(52)	(52)	(52)
Proceeds from investments	480	480	(420)
Other, net	7	(488)	(13)
Total cash used in investing activities	(390)	(1,372)	(740)
Financing activities			
Proceeds from debt	(1,214)	(3,302)	(312)
Repayment of debt	(12)	(46)	(25)
Repayment of debt	(12)	(46)	(25)
Repayment of debt	(12)	(46)	(25)
Other, net	(12)	(46)	(25)
Total cash used in financing activities	(1,238)	(3,840)	(387)
Effect of exchange rate changes on cash	(8)	(2)	7
Net change in cash, cash equivalents and restricted cash	(812)	(1,032)	823
Cash, cash equivalents and restricted cash at beginning of period	\$ 2,636	\$ 3,668	\$ 2,845
Cash, cash equivalents and restricted cash at end of period	\$ 1,824	\$ 2,636	\$ 3,668
Supplemental cash flow information			
Income taxes paid	\$ 264	\$ 718	\$ 442
Interest paid	\$ 31	\$ 22	\$ 50

See Notes to Consolidated Financial Statements

42

Citation chip opens the page

Filing Lens

Local 3B · cited · private

0 doc bytes sent

Test mode: replays recorded responses.

How much cash did operating activities provide in fiscal 2023?

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Confidence 99% Well supported

Calibrated probability the retrieved pages support an answer. Abstains below 50%.

Recorded responses (hand-authored test fixture; replace with Kaggle-recorded Qwen2.5-3B outputs) · 21 ms · citation coverage 100% · sources: p.42

What was Walmart's capital expenditure in fiscal 2023?

I can't answer that from this filing

The retrieved pages don't look like they answer this, so no answer was generated.

Confidence 10% Not supported

Calibrated probability the retrieved pages support an answer. Abstains below 50%.

Closest pages, if you want to check yourself:

p.30 p.29

Ask about this filing... Ask

Unsupported question → abstain

Recorded in headless Chromium at Pixel-7 size (no phone on the build machine). The phone video is [\[ASK: phone recording link\]](#).

Privacy proof

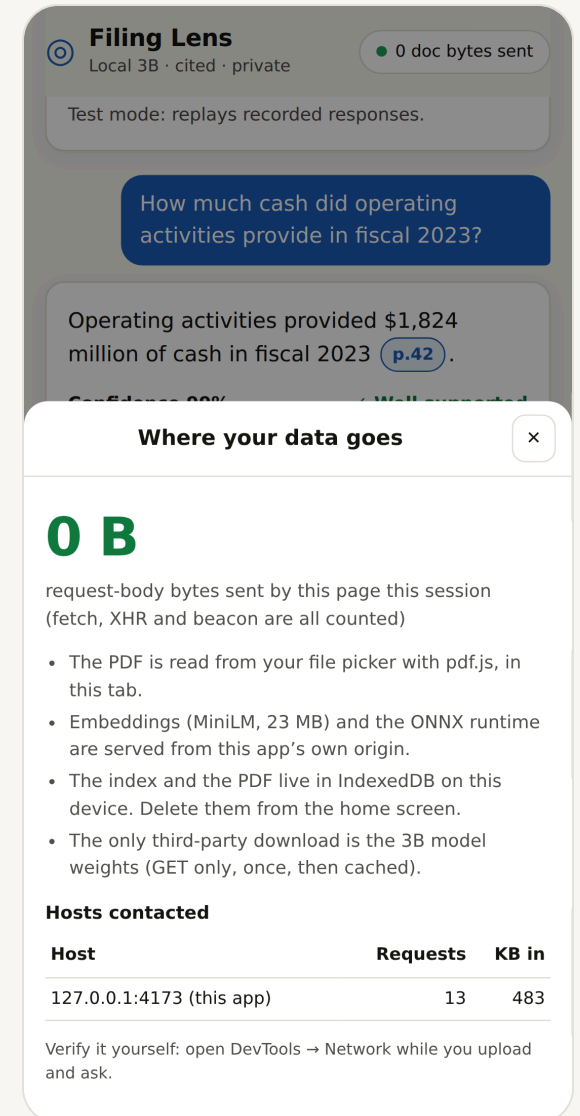
The document never leaves the device. This is enforced by the build's e2e test, not just claimed.

Requests during upload → index → ask → abstain **20**

to any host other than the app's own origin **0**

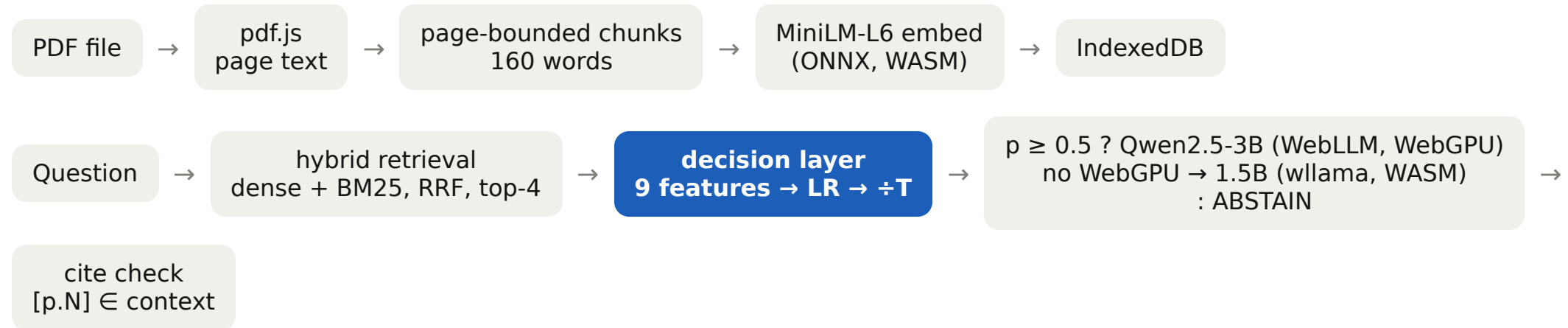
carrying a request body (POST/PUT/beacon) **0**

- pdf.js, MiniLM embeddings (23 MB), ONNX runtime: served from the app origin
- Index + PDF stored in IndexedDB
- Only 3rd-party fetch: the Qwen2.5 weights, GET, once, then cached. After that it works in airplane mode **[ASK: confirm on phone]**
- In-app counter wraps fetch/XHR/beacon and shows request-body bytes: 0



Architecture

On the phone (browser, PWA)

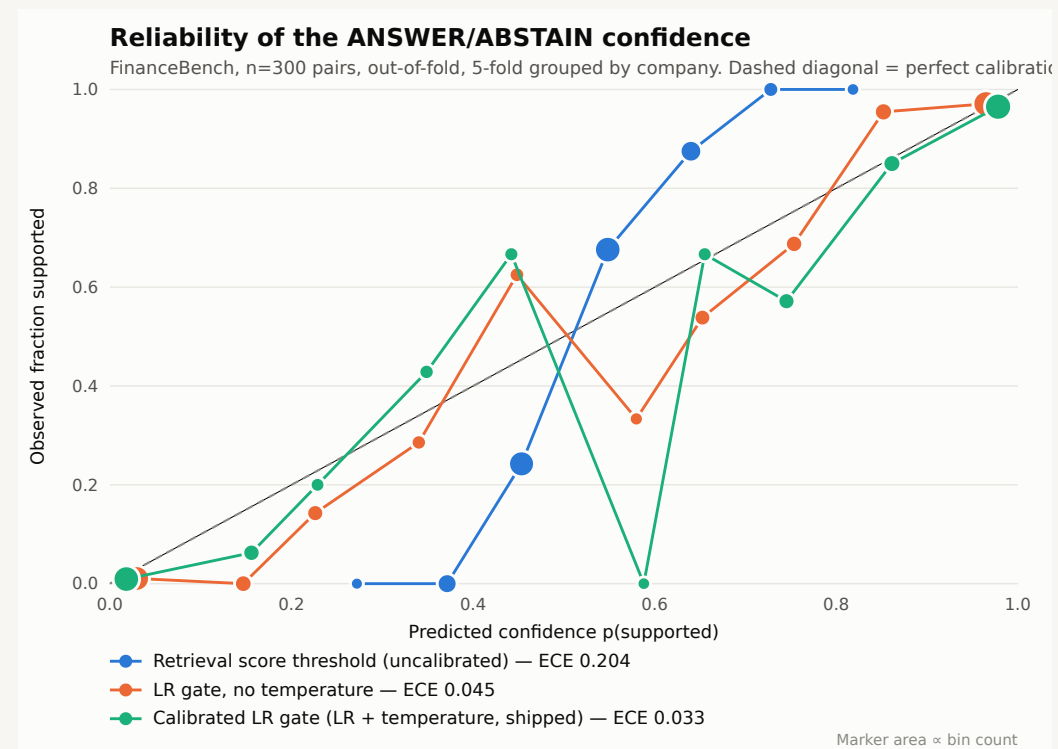


Answer only if (1) calibrated $p(\text{supported}) \geq 0.5$, (2) the model doesn't say NOT IN DOCUMENT, (3) every citation points at a retrieved page and $\geq 50\%$ of sentences are cited. Otherwise show the abstain card with the closest pages.

Bench: decision layer

FinanceBench open-source (patronus-ai/financebench), 150 questions, 84 filings. 300 question-filing pairs, 1:1: 150 gold-answer (own filing), 150 gold-abstain (another company's 10-K). 5-fold CV grouped by company. Gold-page recall: @1 0.23 · @2 0.33 · @4 0.44 · @8 0.63 · @16 0.73.

Gate	Acc.	ECE ↓	Abstain prec.	Abstain rec.	Latency	Cost/q
No gate (always answer)	0.500	0.500	—	0.000	0 ms	\$0
Retrieval score threshold (uncalibrated)	0.780	0.204	0.804	0.740	<0.01 ms	\$0
LR gate, no temperature	0.910	0.045	0.930	0.887	0.28 ms	\$0
Calibrated LR gate (LR + temperature, shipped)	0.910	0.033	0.930	0.887	0.28 ms	\$0
LLM-router (3B YES/NO)	pending (Kaggle)					
Jev	[ASK: what is Jev + can it run locally]					



3B generation (Kaggle T4): citation accuracy pending · answer accuracy pending · tokens/s pending. Retrieval p50 9.5 ms on Node v22.22.0, 4 vCPU container, no GPU. Phone latency **[ASK: measure on iQOO]** .

Limits (honest)

- The 3B was **not run** on the build machine (model hosts blocked). Generation metrics come from the Kaggle T4 run: pending.
- Abstain cases are another company's 10-K: the gate mostly has to notice the question's company isn't in the filing. A near-miss question on the right filing is harder and not in this bench.
- Answer grading is automatic (number within 1%, else token-F1), an approximation of FinanceBench's human grading.
- Scanned PDFs (no text layer) are rejected; no OCR yet. Tables are read as flattened text.
- ~2 GB one-time model download; WebGPU needs ~2.5 GB of GPU memory (WebLLM's figure for Qwen2.5-3B q4f16).

Roadmap

- QA fine-tune of the local 3B on filing questions, if the Kaggle numbers show the gap is generation rather than retrieval
- Train the decision layer on hard negatives: same filing, a question the filing doesn't answer
- Table-aware chunking for financial statements (row/column headers kept with numbers)
- Indian filings: BSE/NSE annual reports, Hindi questions
- Multi-document compare: "this year vs last year" across two 10-Ks, still on-device

The ask

[ASK: what you want from the judges / iQOO: e.g. device lab time on iQOO phones, mentorship, pilot partner]

Repo: github.com/rajyyug1132/Filing-Lens-Private-Cited-Answers

Cited answers, or an honest "I can't tell". Your filing stays yours.